

Rodrigo Alvarez Lucendo

+41 77 289 46 00 | r.alvarezlucendo16@gmail.com | Spanish (EU) citizen | Swiss B permit valid to 2031

Computer scientist with a 30 EC econometrics minor (time series, empirical finance), production ML and data engineering experience at Amazon and an early-stage startup, and an ICML 2026 publication. Educated and employed across Spain, the US, the Netherlands, Luxembourg and Switzerland.

EDUCATION

2023–2026	MSc Computer Science , TU Delft	Netherlands & Switzerland
	- Graduated May 2026; exchange semester and MSc thesis at EPFL, Switzerland. - Courses: Deep Learning, Modern NLP, Graph ML, Deep RL, Distributed ML Systems.	
2022–2023	Minor in Applied Econometrics (30 EC) , Vrije Universiteit Amsterdam	Netherlands
	- Time series econometrics, empirical finance, empirical economics. - Capstone: real-life econometric and data-science modelling case study on market data.	
2019–2023	BSc Computer Science , TU Delft	Netherlands
	Thesis: spatiotemporal deep-learning forecasting model for algal blooms, trained on HPC.	

WORK EXPERIENCE

Jun 2025–Present	Machine Learning Engineer , Xcert.ai	Lausanne, Switzerland
	- First employee at a CHF 1.5M+ funded startup building AI assistants for aerospace certification. - Led RAG pipeline development (parsing, chunking, retrieval, evaluation) and built, monitored and evaluated an agentic workflow using the OpenAI Agents SDK. - Shipped to production on FastAPI, Postgres, Neo4j and Docker.	
Jul 2024–Jan 2025	Data Engineer Intern , Amazon	Luxembourg
	- Deployed a monitoring tool for machine learning models running in production. - First in team to explore AI agents; built a ReAct agent for natural-language metric analysis. - Delivered complex SQL queries and business analysis to BI teams.	
2022–2024	Teaching Assistant , TU Delft & EPFL	Netherlands & Switzerland
	Taught 300+ BSc/MSc students (ML, Big Data, Deep Learning); Head TA, Computational Intelligence.	

RESEARCH & PUBLICATIONS

Incremental Learning of Sparse Attention Patterns in Transformers. Oğuz Kaan Yüksel, **Rodrigo Alvarez Lucendo**, Nicolas Flammarion. Second author; accepted to the **ICML 2026** (International Conference on Machine Learning) main conference — one of the world’s top three peer-reviewed AI conferences. Earlier version at the EurIPS 2025 PriGM Workshop. arXiv:2602.19143.

2025–2026	Master Thesis , Natural Language Processing Lab, EPFL	Switzerland
	Measuring scaling laws for memorization in LLMs across model size and data composition, training GPT-2 models from scratch on synthetic and natural data.	
Feb–Sep 2025	Research Assistant , Theory of Machine Learning Lab, EPFL	Switzerland
	Investigated how Transformers learn high-order Markov chains through staged dynamics.	

PROJECTS

nanoGPT: Built nanoGPT from scratch (decoder-only Transformer, BPE tokenizer) and extended it with MoE.
FETCH, EPFL AI Team: Co-developed a natural-language mission planner for a Unitree Go2 robot using LLMs and live sensor input; integrated Nav2 and MCP servers.
AI Tutor: Fine-tuned a 0.6B LLM with SFT and DPO (Hugging Face) to create an AI STEM tutor.

SKILLS

Programming	Python, SQL, Java, Scala, TypeScript, Bash.
Quantitative	Time series econometrics, forecasting, empirical finance, statistical modelling, deep learning.
ML / Data	PyTorch, Hugging Face, PyTorch Lightning, LlamaIndex, Weights & Biases.
Systems / Tools	AWS, Docker, FastAPI, Redis, Neo4j, Git, HPC.
Languages	Spanish (native), English (fluent), French (learning).
Commodities	Self-study: <i>Commodities Demystified</i> (Trafigura), <i>The World for Sale</i> (Blas & Farchy).
Interests	Rowing, alpine skiing, strength training.